

13. Write a Pandas program to detect missing values of a given DataFrame. Display True or False. give simple python code

CODE:

```
import pandas as pd

import numpy as np

df = pd.DataFrame({

    'ord_no': ['70001.0', np.nan, '70002.0', '70004.0', np.nan, '70005.0', '--', '70010.0',
'70003.0', '70012.0', np.nan, '70013.0'],

    'purch_amt': [150.50, 270.65, 65.26, 110.50, 948.50, 2400.60, 5760.00, '?', 12.43,
2480.40, 250.45, 3045.60],

    'ord_date': ['?', '2012-09-10', np.nan, '2012-08-17', '2012-09-10', '2012-07-27', '2012-09-
10', '2012-10-10', '2012-10-10', '2012-06-27', '2012-08-17', '2012-04-25'],

    'customer_id': [3002, 3001, 3001, 3003, 3002, 3001, 3001, 3004, '--', 3002, 3001,
3001],

    'salesman_id': [5002, 5003, '?', 5001, np.nan, 5002, 5001, '?', 5003, 5002, 5003, '--']

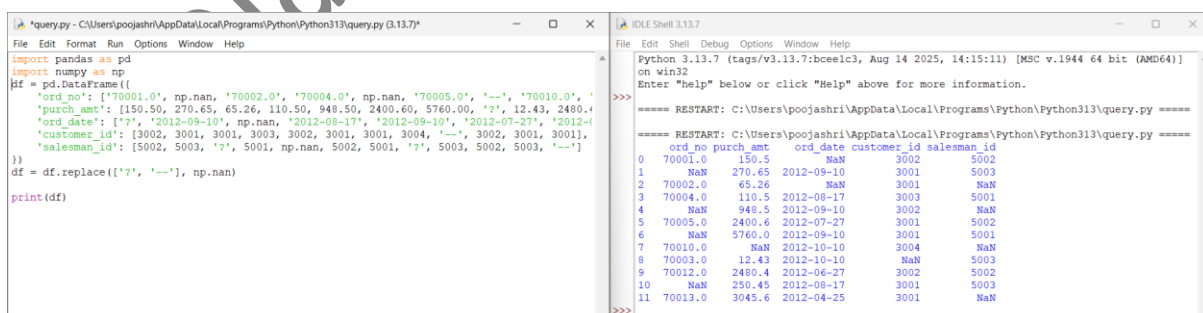
})

# Replace values with no useful information ('?', '--') with NaN

df = df.replace(['?', '--'], np.nan)
```

print(df)

Output:



```
File Edit Format Run Options Window Help
query.py - C:\Users\poojashri\AppData\Local\Programs\Python\Python313\query.py (3.13.7)

import pandas as pd
import numpy as np
df = pd.DataFrame({
    'ord_no': ['70001.0', np.nan, '70002.0', '70004.0', np.nan, '70005.0', '--', '70010.0',
'70003.0', '70012.0', np.nan, '70013.0'],
    'purch_amt': [150.50, 270.65, 65.26, 110.50, 948.50, 2400.60, 5760.00, '?', 12.43,
2480.40, 250.45, 3045.60],
    'ord_date': ['?', '2012-09-10', np.nan, '2012-08-17', '2012-09-10', '2012-07-27', '2012-09-
10', '2012-10-10', '2012-10-10', '2012-06-27', '2012-08-17', '2012-04-25'],
    'customer_id': [3002, 3001, 3001, 3003, 3002, 3001, 3001, 3004, '--', 3002, 3001,
3001],
    'salesman_id': [5002, 5003, '?', 5001, np.nan, 5002, 5001, '?', 5003, 5002, 5003, '--']
})
df = df.replace(['?', '--'], np.nan)
print(df)

Python 3.13.7 (tags/v3.13.7:bce1c3, Aug 14 2025, 14:15:11) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.

===== RESTART: C:\Users\poojashri\AppData\Local\Programs\Python\Python313\query.py =====
===== RESTART: C:\Users\poojashri\AppData\Local\Programs\Python\Python313\query.py =====
   ord_no  purch_amt  ord_date  customer_id  salesman_id
0  70001.0    150.50      NaN           3002           5002
1      NaN    270.65  2012-09-10           3001           5003
2  70002.0     65.26      NaN           3001           NaN
3  70004.0    110.50  2012-08-17           3003           5001
4      NaN    948.50  2012-09-10           3002           NaN
5  70005.0   2400.60  2012-07-27           3001           5002
6      NaN    5760.00  2012-09-10           3001           5001
7  70010.0      NaN  2012-10-10           3004           NaN
8  70003.0     12.43  2012-10-10           NaN           5003
9  70012.0   2480.40  2012-06-27           3002           5002
10      NaN    250.45  2012-08-17           3001           5003
11  70013.0   3045.60  2012-04-25           3001           NaN
```

14. Write a Pandas program to find and replace the missing values in a given DataFrame which do not have any valuable information.

Code:

```

import pandas as pd

import numpy as np

df = pd.DataFrame({

    'ord_no':
[70001,np.nan,70002,70004,np.nan,70005,np.nan,70010,70003,70012,np.nan,70013],

    'purch_amt':
[150.50,270.65,65.26,110.50,948.50,2400.60,5760.00,1983.43,2480.40,250.45,75.29,3
045.60],

    'ord_date': ['2012-10-05','2012-09-10',np.nan,'2012-08-17','2012-09-10','2012-07-
27','2012-09-10','2012-10-10','2012-10-10','2012-06-27','2012-08-17','2012-04-25'],

    'customer_id': [3002,3001,3001,3003,3002,3001,3001,3004,3003,3002,3001,3001],

    'salesman_id':
[5002.0,5003.0,5001.0,np.nan,5002.0,5001.0,5001.0,np.nan,5003.0,5002.0,5003.0,np.
nan]

})

print(df.isnull())

```

Output:

```

query.py - C:\Users\poojashri\AppData\Local\Programs\Python\Python313\query.py (3.13.7)
File Edit Format Run Options Window Help

import pandas as pd
import numpy as np

df = pd.DataFrame({
    'ord_no': [70001,np.nan,70002,70004,np.nan,70005,np.nan,70010,70003,70012,np.nan,70013],
    'purch_amt': [150.50,270.65,65.26,110.50,948.50,2400.60,5760.00,1983.43,2480.40,250.45,75.29,3045.60],
    'ord_date': ['2012-10-05','2012-09-10',np.nan,'2012-08-17','2012-09-10','2012-07-27','2012-09-10','2012-10-10','2012-10-10','2012-06-27','2012-08-17','2012-04-25'],
    'customer_id': [3002,3001,3001,3003,3002,3001,3001,3004,3003,3002,3001,3001],
    'salesman_id': [5002.0,5003.0,5001.0,np.nan,5002.0,5001.0,5001.0,np.nan,5003.0,5002.0,5003.0,np.nan]
})

print(df.isnull())

```

```

IDLE Shell 3.13.7
Python 3.13.7 (tags/v3.13.7:bce1c3, Aug 14 2025, 14:15:11) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.

>>>
* RESTART: C:\Users\poojashri\AppData\Local\Programs\Python\Python313\query.py =
ord_no purch_amt ord_date customer_id salesman_id
0 False False False False False
1 True False False False False
2 False False True False False
3 False False False False True
4 True False False False False
5 False False False False False
6 True False False False False
7 False False False False True
8 False False False False False
9 False False False False False
10 True False False False False
11 False False False False True
>>>

```